Q. 1. Hollow Square.

```
*  *  *  *  *
*  _  _  _  *
*  _  _  _  *
*  _  _  _  *
*  *  *  *  *
```

- code

```
int n = sc.nextInt ();
for (int i=1; i<= n; i++){
    for (int j=1; j<= n; j++)
        if (2 <= i <= 4 && 2<=j<=
            sop(" ");
    }
    else{
        sop("* _ ");
    }
    sop(" ");
}
```

Dry Run -

i<=n

i<=5 j<=n

↳ j<=5

↳ *

2<=5

↳ *

3<=5

↳ *

4<=5

↳ *

5<=5

↳ *

2 <=5

↳ 1<=5

↳ *

2 <=5

↳ *

3 <=5

↳ _

4 <=5

↳ _

5 <=5

↳ *

Q.2.

```
*     *
  *   *
    *
  *   *
*     *
```

→
```
import java.util.Scanner;
class Demo{
        psvm(){
            Scanner sc = new Scanner(System.in);
            int n = sc.nextInt();
            for(int i = 1; i<=n; i++){
                for(int j=1; j<=n; j++){
                    if(i==j || i+j==n+1){
                        sop("*");
                    }
                    else{
                        sop(" ");
                    }
                }
                sop();
            }
        }
}
```

total no of our rows / colum

2==6

- Dry Run --

~~n = 5~~ n = 5

i = 1, i<=n, i++

1 <= 5

↳ 1 <= 5

↳ (i=j || i+j = 6)

↳ *

2 <= 5

↳ 0

3 <= 5

↳ 0

4 <= 5

↳ 0

5 <= 5

↳ * (i=j || i+j=n+1)

Q.3

```
*          → star ekda print hota hota   [1→i]   [1,1]
* *        → star 2 vela                 [1→i]   [1,2]
* * *      → 3 vela                      [1→i]   [1,3]
* * * *    → 4 vela                      -||-    -||-
* * * * *  → 5 vela                      -||-    -||-
```

→ import java.util.scanner;

```
class Demo {
        psvm (){
            Scanner sc = new Scanner (system.in);
                sop("enter no. of rows : ");
                int n = sc.nextInt ();
            for(int i=1; i<=n; i++){
                for(int j=1; j<=i; j++){
                    sop("*_");
                }
                sop();
            }
        }
}
```

Dry Run -

```
1 <= 5
  → j <= i
    1 <= 1
      → 1 <= 1
        → *
      2 <= 2
        → 0
    1 <= 2
      → 1 <= 2
        → *
      2 <= 2
        → *
      3 <= 2
        → 0
```

Q. 4.

```
* * * * *
* * * *
* * *
* *
*
```

→

```
import java.util.Scanner;
class Demo{
        psvm (){
            Scanner sc = new Scanner (System.in);
                sop("enter no. of rows: ");
                int n = sc.nextInt();
                sop("enter no. of count : ");
                int count = sc.nextInt().
                for(int i=1; i<=n; i++){
                    for(int j=1; j<=count; j++){
                        sop("* ");
                    }
                    count --;
                    sop( );
                }
        }
}
```

or

```
for(int i=1; i≤n; i++){
    for(int j=1; j≤n-j+1; j++){
        sop("* ");
    }
    sop();
}
```

- Dry Run.

$\boxed{n-j+1}$

Rows ↓ increase

stars ↓ decrease

Q.5

```
* # * #
# * # *
* # * #
# * # *
```

→
```
import java.util.scanner;
class Demo{
        psvm(){
            Scanner sc = new scanner(system.in);
                sop(" enter no. of rows : ");
                int n = sc.nextInt();
                for (int i=1; i<=n; j++){
                    for (int j=1; j<=n; j++){
                        if ((i+j)%2 ==0){
                            sop("* ");
                        }
                        else{
                            sop("# ");
                        }
                    }sop( );
                }
            }
        }
```

→ Dry Run.

1 <= 4

↳ 1 <= 4 j++

↳ if ((i+j)%2 ==0){

sop (*)

}

↳ *

2 <= 4

↳ #

Q.6.

```
        *
      * *
    * * *
  * * * *
* * * * *
```

```
n = 5
for (int i = 1; i <= 5; i++){
    for (int j = 1; j <= 5; j++){
        if (j >= n - i + 1){
            print("* ");
        }
        else{
            print(" ");
        }
    }
}
```

Dry Run.

$1 <= 5$
↳ $j <= n - i + 1$
print(*)

$2 <= 5$
↳ $4 = 5 - 2 + 1$

$1 <= 5$
↳ $1 <= 5$
↳ $j = = 5 - 1 + 1$

Q.7.

```
* * * * *
  * * * *
    * * *
      * *
        *
```

$5 <= 1$
↳ $j <= 5$
↳ $i > j$
↳

```
if (i > j){
    print(-)
}
else{
    (*)
}.
```